

Latent Space Oddity: Riemannian Geometry in VAEs

Unifying Interpolation and Sampling via Geometry

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December 17, 2025

The "Flat Earth" Assumption of VAEs

How Standard VAEs View Latent Space:

- The latent space $\mathcal{Z} = \mathbb{R}^d$ is treated as a **flat Euclidean space**.
- Distances are measured via the standard L_2 norm:

$$d(z_1, z_2) = \|z_1 - z_2\|_2$$

- **Interpolation Strategy:** Linear paths are assumed to be optimal.

$$z(t) = (1 - t)z_0 + tz_1$$

The Reality: Latent Space is Curved

The Reality Gap:

- The decoder $f_\theta : \mathcal{Z} \rightarrow \mathcal{X}$ is a non-linear mapping (neural network).
- It defines a curved **manifold** embedded in the data space.
- **Consequence:** A straight line in $\mathcal{Z}$ bends and twists when mapped to $\mathcal{X}$.

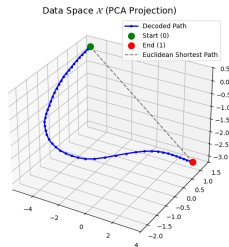
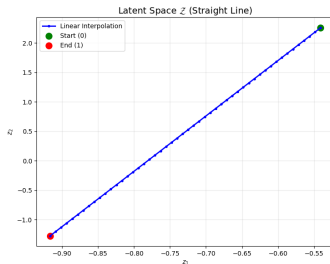


Figure: **Left:** Linear path in Latent Space (Blue line). **Right:** The actual curved path taken in Data Space (PCA projection).

VAE: The Probabilistic Framework

Before fixing the geometry, let's recall the architecture.

1. Encoder (Inference)

- Maps input x to latent posterior approx.
- $q_{\phi}(z|x) = \mathcal{N}(z | \mu_{\phi}(x), \text{diag}(\sigma_{\phi}^2(x)))$
- *Role:* Compresses data into codes.

2. Decoder (Generator)

- Maps latent z back to data space.
- $p_{\theta}(x|z) = \mathcal{N}(x | \mu_{\theta}(z), \Sigma_{\theta}(z))$
- *Role:* Defines the data manifold.

Optimization Objective (ELBO):

$$\mathcal{L}_{\text{VAE}} = \underbrace{-\mathbb{E}_{q_{\phi}}[\log p_{\theta}(x|z)]}_{\text{Reconstruction (MSE)}} + \beta \cdot \underbrace{D_{\text{KL}}(q_{\phi}(z|x) || \mathcal{N}(0, I))}_{\text{Regularization (Euclidean)}}$$

Riemannian Metric

The decoder $f_\theta : \mathcal{Z} \rightarrow \mathcal{X}$ immerses the latent space into the data space. We can "pull back" the geometry of the data space into $\mathcal{Z}$.

The Pull-back Metric Tensor

For a generator $f_\theta(z)$, the Riemannian metric $M_z \in \mathbb{R}^{d \times d}$ is given by:

$$M_z = \mathbf{J}_f(z)^\top \mathbf{J}_f(z)$$

where $\mathbf{J}_f(z) = \frac{\partial f}{\partial z}$ is the Jacobian matrix of the decoder.

Physical Meaning: It measures **local sensitivity**. A small step dz in latent space results in a change in data space of squared length:

$$ds^2 = dz^\top M_z dz$$

Interpreting the Metric Landscape

- **High Curvature** ($\det M_z$ large):
 - A small change in z causes a *large* change in the image x .
 - Corresponds to **data-rich regions** (the manifold).
- **Low Curvature** ($\det M_z \approx 0$):
 - Changing z barely affects the output image.
 - Corresponds to **empty space** or "dead zones."

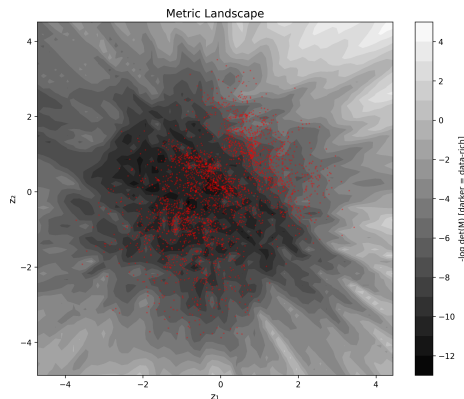


Figure: High curvature regions = data-rich. Light regions = empty space.

RBF Variance Network

Motivation:

- Standard Neural Networks extrapolate linearly in empty space.
- **Risk:** The metric M_z might remain small in "dead zones," allowing geodesics to cross them.

Solution: We parameterize the precision $\beta(z)$ as a mixture of radial basis functions, similar to a **Gaussian Mixture Model (GMM)** fixed on the training data.

$$\beta(z) = \Sigma_{\theta}(z)^{-1} = \sum_{k=1}^K w_k \exp\left(-\frac{\|z - c_k\|^2}{2\lambda^2}\right)$$

Geometric Effect:

- **Near Data (c_k):** Variance is low $\rightarrow$ Metric is normal.
- **Far from Data:** Variance explodes ($\beta \rightarrow 0$) $\rightarrow$ Metric explodes.

RBF Coverage Example

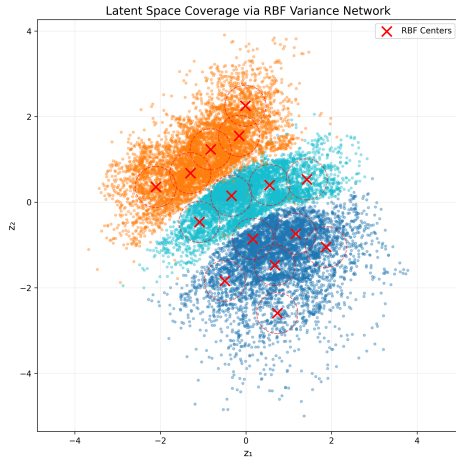


Figure: RBF centers (red X) and coverage circles. Sparse zones get larger variance, forcing trajectories to stay near data.

Definition: A geodesic $\gamma(t)$ minimizes the energy functional:

$$\mathcal{E}(\gamma) = \int_0^1 \sqrt{\dot{\gamma}(t)^\top M_{\gamma(t)} \dot{\gamma}(t)} dt$$

- Linear paths fail in curved latent spaces.
- Geodesics respect latent geometry.

Geodesic vs Linear Interpolation

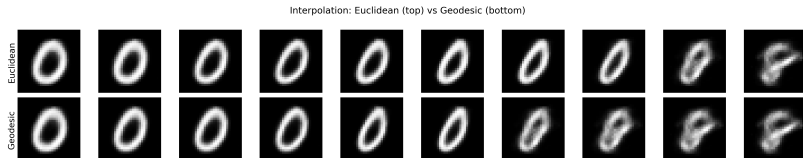


Figure: First Line = Euclidean (linear). Second = Geodesic interpolation. Geodesics stay on high-density manifold.

Beyond Interpolation: Geometry-Aware Sampling

Interpolation fixes paths; geometry-aware sampling captures uncertainty.

Bayesian Goal:

- Sample from the posterior $p(z|x)$ in latent space.
- Capture multimodality and uncertainty beyond Gaussian approximations.

Limitation of Standard MCMC:

- Random-walk Metropolis–Hastings uses isotropic proposals.
- In curved or anisotropic latent spaces:
 - High rejection rate
 - Poor exploration of the manifold

Idea: Adapt proposals using the **local latent geometry and RBF variance network**.

Geometry-Aware Metropolis–Hastings

Key Idea: Replace isotropic Gaussian proposals with geometry-aware ones.

Metric- and RBF-Informed Proposal

At latent position z , propose:

$$z' = z + \epsilon \mathcal{N}(0, \sigma(z)^2)$$

where $\sigma(z)$ is computed from:

- Local Riemannian metric M_z
- RBF variance network trained on latent codes

Interpretation:

- High curvature regions $\Rightarrow$ smaller steps
- Sparse / flat regions $\Rightarrow$ larger steps
- Proposals stay near the data manifold

Acceptance Rule:

$$\alpha = \min \left(1, \frac{p(z'|x) q(z|z')}{p(z|x) q(z'|z)} \right)$$

Adaptive MCMC via Robbins–Monro

Problem: Choosing ϵ for efficient exploration.

Solution: Adaptive MCMC

- Target an optimal acceptance rate (e.g., 23%)
- Update ϵ online using stochastic approximation

Robbins–Monro Update

$$\epsilon_{t+1} = \epsilon_t + \eta_t (\alpha_t - \alpha^*)$$

where:

- α_t = observed acceptance rate
- α^* = target acceptance rate
- $\eta_t = 1/\sqrt{t}$ = decreasing step size

Outcome:

- Automatic adaptation to local geometry
- Improved mixing along the latent manifold
- Stabilizes acceptance near α^* without manual tuning

Adaptive Gibbs Sampling in Latent Space

Comparison of Strategies:

- Vanilla Gibbs: isotropic proposals $\rightarrow$ poor mixing
- Adaptive Gibbs: automatic scale adaptation $\rightarrow$ better acceptance
- Adaptive + RBF: geometry-aware proposals $\rightarrow$ stays on manifold

Diagnostics and Metrics:

- Acceptance rates 23%
- Faster autocorrelation decay
- Trajectories concentrated on high-density manifold

Experimental Results: Trajectory

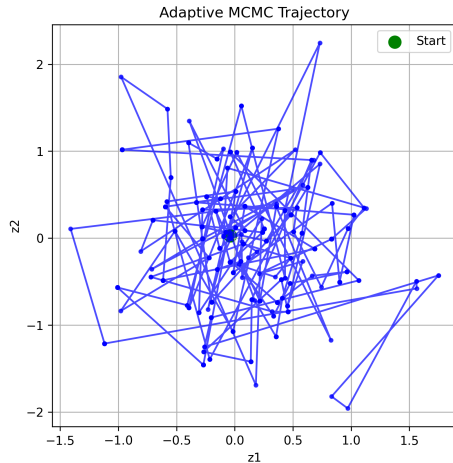


Figure: Adaptive MCMC trajectory starting from a latent point. Paths follow the high-density manifold.

Experimental Results: Acceptance & Decorrelation

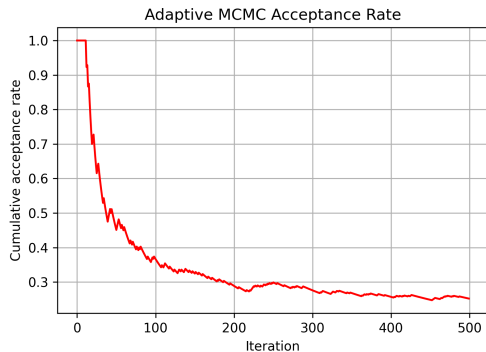


Figure: Acceptance rate adapts to target 23% over iterations.

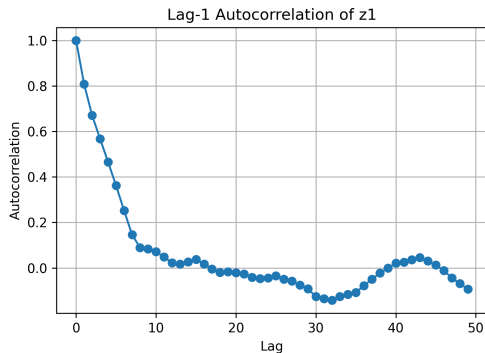


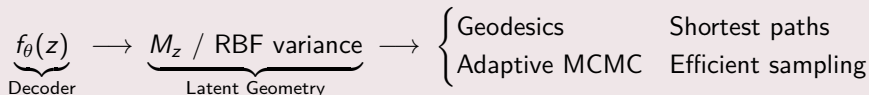
Figure: Autocorrelation decays faster using adaptive geometry-aware proposals.

Conclusion

Take-Home Messages:

- ① Euclidean operations in latent space ignore local curvature.
- ② Decoder defines a *Riemannian metric* capturing local density.
- ③ Geometry-aware methods improve:
 - Interpolation via geodesics
 - Sampling via adaptive, geometry-informed MCMC

From Geometry to Algorithms



Methodological Contributions:

- Metric estimation via decoder Jacobian
- Variance regularization via RBF network
- Adaptive step sizes with stochastic approximation

- Arvanitidis et al., *Latent Space Oddity: on the Curvature of Deep Generative Models*, ICML 2018.
- Kingma Welling, *Auto-Encoding Variational Bayes*, ICLR 2014.
- Rezende et al., *Stochastic Backpropagation and Approximate Inference in Deep Generative Models*, ICML 2014.